

TUBUAI, French Polynesia

WMO: 919540

Lat: 23.371S

Lon: 149.529W

Elev: 20

StdP: 14.69

Time Zone: -10.00 (W10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	59.3	61.5	48.7	50.9	66.6	50.6	54.6	66.9	29.5	70.7	26.4	71.1	3.9	170	0.374

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	7.5	85.2	77.5	84.3	77.1	83.5	76.6	79.1	83.0	78.4	82.3	77.9	81.9	10.9	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.9	145.1	81.7	77.2	141.6	81.2	76.5	138.3	80.8	42.6	83.1	41.9	82.6	41.3	82.0	82.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.1	21.9	19.5	DB	55.7	86.4	1.9	0.8	54.3	87.0	53.2	87.5	52.1	88.0	50.7	88.6	(4)
(5)			WB	53.4	79.9	1.4	1.0	52.3	80.6	51.5	81.2	50.7	81.8	49.6	82.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	74.3	78.2	79.1	78.7	76.7	73.9	71.7	70.4	70.0	70.4	71.9	74.4	76.3		
	DBStd	4.18	2.17	2.05	2.24	2.43	2.84	2.80	3.12	3.00	2.84	2.60	2.57	2.31		
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD65	4	0	0	0	0	0	0	2	1	1	0	0	0		
	CDD50	8862	874	815	889	802	741	650	632	619	613	680	731	816		
	CDD65	3392	409	395	424	352	277	200	169	155	164	215	281	351		
	CDH74	18505	3261	3483	3605	2270	1094	415	247	199	239	439	1197	2056		
	CDH80	2065	429	614	610	207	24	1	0	2	2	37	139			
(14) Wind	WSAvg	10.1	10.1	10.4	9.3	10.0	8.4	10.1	10.5	11.0	10.8	10.3	10.2	10.4		
(15) Precipitation	PrecAvg	73.3	7.4	7.9	7.4	6.9	5.8	4.6	5.4	5.4	4.1	4.3	5.2	8.1		
	PrecMax	111.8	21.1	22.9	22.6	17.6	17.1	13.0	11.8	12.8	13.0	11.7	13.6	23.7		
	PrecMin	46.9	0.8	1.0	0.9	1.2	0.4	0.3	1.4	1.6	0.4	0.4	0.3	0.6		
	PrecStd	13.8	4.1	4.6	4.6	3.7	3.5	2.5	2.7	2.9	3.1	2.6	2.9	4.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	85.4	86.4	86.3	84.6	81.9	80.0	78.9	78.7	79.8	80.1	82.6	84.3		
		MCWB	77.4	78.1	78.4	77.3	76.0	74.8	73.8	72.6	73.4	73.8	75.3	77.5		
	2%	DB	84.3	85.2	85.0	83.4	80.6	78.5	77.4	77.3	77.9	78.6	81.0	82.7		
		MCWB	76.8	77.5	77.6	76.6	75.0	73.3	72.3	72.0	71.7	72.4	74.2	76.5		
	5%	DB	83.3	84.3	84.1	82.2	79.5	77.4	76.4	76.1	76.5	77.4	80.0	81.6		
		MCWB	76.4	77.0	77.0	75.8	74.2	72.5	71.5	71.1	70.8	71.4	73.7	75.7		
	10%	DB	82.4	83.4	83.1	81.1	78.5	76.4	75.4	75.1	75.2	76.4	79.0	80.5		
		MCWB	75.9	76.6	76.6	75.4	73.5	71.6	70.9	70.2	69.8	70.6	73.2	74.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	79.2	79.9	79.9	79.2	77.6	76.1	75.3	74.6	75.1	75.8	77.2	79.0		
		MCDB	82.8	84.2	84.0	82.5	80.4	78.6	77.5	76.8	78.1	78.4	80.2	82.0		
	2%	WB	78.3	78.8	79.0	78.2	76.7	74.9	73.9	73.6	73.6	74.4	76.2	77.9		
		MCDB	82.0	83.3	83.2	81.7	79.3	77.5	76.3	76.1	76.1	77.1	79.2	81.1		
	5%	WB	77.7	78.1	78.3	77.3	75.7	73.9	73.1	72.6	72.5	73.4	75.4	77.0		
		MCDB	81.5	82.6	82.6	81.0	78.6	76.6	75.6	75.3	75.3	76.1	78.6	80.4		
	10%	WB	77.0	77.4	77.6	76.4	74.7	72.7	71.9	71.5	71.1	72.3	74.5	76.1		
		MCDB	81.1	82.0	82.0	80.3	77.8	75.7	74.8	74.4	74.4	75.4	77.9	79.5		
(35) Mean Daily Temperature Range	MDBR		7.3	7.5	8.0	7.3	7.7	7.6	8.0	8.0	7.8	7.8	7.6	7.1		
	5% DB	MCDBR	8.4	8.6	8.8	8.4	8.4	8.4	8.3	8.4	8.8	8.5	8.4	8.1		
		MCWBR	4.2	4.2	4.4	4.6	5.0	5.3	5.4	5.5	5.3	4.9	4.5	4.3		
	5% WB	MCDBR	7.1	7.6	7.7	7.4	6.9	7.0	7.1	7.2	7.8	7.1	6.9	7.0		
		MCWBR	4.1	4.2	4.3	4.7	4.9	5.3	5.5	5.7	5.9	4.9	4.4	4.4		
(40) Clear-Sky Solar Irradiance	taub		0.377	0.381	0.372	0.378	0.367	0.358	0.353	0.355	0.369	0.365	0.370	0.375		
	taud		2.625	2.615	2.646	2.625	2.631	2.639	2.639	2.625	2.576	2.581	2.601	2.618		
	Ebn at Noon		307	301	296	280	271	267	272	283	292	302	307	307		
	Edh at Noon		32	32	30	28	26	25	25	28	31	33	33	33		
(44) All-Sky Solar Radiation	RadAvg		2052	1926	1736	1382	1155	1007	1123	1325	1647	1878	2034	1915		
	RadStd		206	202	173	108	96	68	92	98	127	106	167	170		

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	+1.39	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page